

CASMA — Computer-Aided Sperm Morphology Analyser

Diagnostic-grade image analysis tool for quantifying sperm morphology from microscopy images against WHO 6th Edition strict criteria.

Overview

CASMA is a Python-based image analysis pipeline designed to replicate the workflow of commercial computer-aided sperm analysis (CASA/CASMA) systems used in UKAS-accredited andrology laboratories. It segments sperm heads from brightfield microscopy images, computes six morphometric parameters, classifies each sperm as normal or abnormal against WHO reference ranges, and generates ALCOA+-compliant audit trails for diagnostic quality assurance.

Built as part of independent research into diagnostic andrology workflows and laboratory data integrity.

Features

Module	Description
Image Preprocessing	Gaussian denoising, adaptive Otsu thresholding, morphological opening/closing, hole filling
Segmentation	Connected-component labelling with size-based filtering to isolate sperm heads from debris
Morphometrics	Principal-component analysis (inertia tensor) for major/minor axis estimation; area, perimeter, circularity, aspect ratio
WHO Classification	Strict morphology criteria from <i>WHO Laboratory Manual for the Examination and Processing of Human Semen</i> (6th Edition, 2021)
Robust QC	Modified Z-score outlier detection using Median Absolute Deviation (MAD) —robust to small sample sizes
Visualisation	Annotated microscopy overlays (green = normal, red = abnormal) + distribution histograms with WHO reference lines
Reporting	Per-sperm CSV, population summary JSON, and ALCOA+ compliant audit trail

WHO 6th Edition Criteria Applied

Parameter	Normal Range	Source
Head length	4.0 –5.0 µm	WHO 6th Ed., Table 2.5
Head width	2.5 –3.5 µm	WHO 6th Ed., Table 2.5
Head area	12.0 –22.0 µm²	Derived from length × width
Perimeter	13.0 –19.0 µm	Approximated from contour
Circularity	0.75	Roundness index
Aspect ratio	1.20 –1.80	Length / width

Table 2 – continued

Parameter	Normal Range	Source
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Lower reference limit for normal morphology: 4.0%

Installation

```
git clone https://github.com/AdrianHall23/CASMA-Sperm-Morphology-Analyser.git
cd CASMA-Sperm-Morphology-Analyser
pip install -r requirements.txt
```

Requirements

- Python 3.9
- numpy
- scipy
- matplotlib
- Pillow

Usage

Basic run

```
python casma_analyser.py --image sample_image.tif --scale 0.12
```

Arguments

Flag	Description	Default
--image	Path to microscopy image (TIFF, PNG, JPG)	<i>required</i>
--scale	Micrometres per pixel (calibrate per objective)	0.12
--outdir	Output directory for reports	casma_output

Example with 40× objective

```
python casma_analyser.py --image patient_sample_001.tif --scale 0.12 --outdir results/patient_001
```

Outputs

```
casma_output/
├─ annotated_image.png      # Original image with measured sperm highlighted
├─ morphology_distributions.png # Histograms of all 6 parameters vs WHO ranges
```

```
|— morphology_report.csv      # Per-sperm morphometrics + normal/abnormal flag
|— summary_statistics.json    # Population stats: mean, std, median, normal %
|— qc_audit_trail.txt         # ALCOA+ compliant audit log
```

Sample Results

Annotated Microscopy Image

Green dots = normal morphology; Red dots = abnormal morphology. Numbers correspond to rows in CSV report.

Morphometric Distributions

Population distributions with WHO 6th Edition reference range boundaries (dashed green lines).

Methodology

1. Segmentation

Sperm heads are isolated using adaptive thresholding followed by morphological operations. Connected components are filtered by area (80–800 px) to exclude debris and overlapping cells.

2. Principal Axes Estimation

Rather than using simple bounding boxes, CASMA computes the **inertia tensor** (covariance matrix of pixel coordinates) and extracts eigenvalues. The square roots of the eigenvalues, scaled by 4, provide robust estimates of the major and minor axes — this method is invariant to rotation and more accurate than rectangular bounding boxes for elliptical objects.

3. Robust Outlier Detection

Standard Z-scores assume normality and are unreliable for small samples ($n < 30$). CASMA uses the **modified Z-score** with Median Absolute Deviation (MAD):

$$MZ_i = 0.6745 \times (x_i - \text{median}) / \text{MAD}$$

Where $\text{MAD} = \text{median}(|x_i - \text{median}|)$. Values with $|MZ_i| > 2.5$ are flagged as population outliers.

4. ALCOA+ Compliance

Every run generates an audit trail documenting:

- **Attributable:** Automated pipeline with version-tagged source code
 - **Legible:** Standard CSV/JSON/PNG outputs
 - **Contemporaneous:** ISO 8601 timestamp at runtime
 - **Original:** Raw image preserved; all calculations reproducible
 - **Accurate:** WHO 6th Edition reference ranges hard-coded and versioned
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Limitations & Future Work

- **Brightfield only:** Currently tested on phase-contrast and brightfield images. Fluorescence support planned.
 - **2D analysis:** Does not measure midpiece or tail morphology (requires multi-focus or AI segmentation).
 - **Single frame:** Batch processing of time-lapse sequences is on the roadmap.
 - **Validation:** Pending correlation study against manual morphology assessment by trained andrologists.
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References

1. World Health Organization. *WHO Laboratory Manual for the Examination and Processing of Human Semen* (6th ed.). Geneva: WHO Press; 2021.
 2. Iglewicz B, Hoaglin DC. *How to Detect and Handle Outliers*. Milwaukee: ASQC Quality Press; 1993. (Modified Z-score / MAD method)
 3. Young IT. *Proof without prejudice: use of the Kolmogorov-Smirnov test for the analysis of histograms from flow systems and other sources*. J Histochem Cytochem. 1977;25(7):935–941.
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